Thomas Malthus and … Elon Musk?

Lecture 1.5

2/3/26

Premodern population beliefs

High fertility + High mortality -- “Childhood” didn’t really exist

Enlightenment

World population ~doubled from 1500-1800 after being steady for a long time

Reduction in child mortality, recognition of children + childhood as special category of humanity

Thomas Malthus

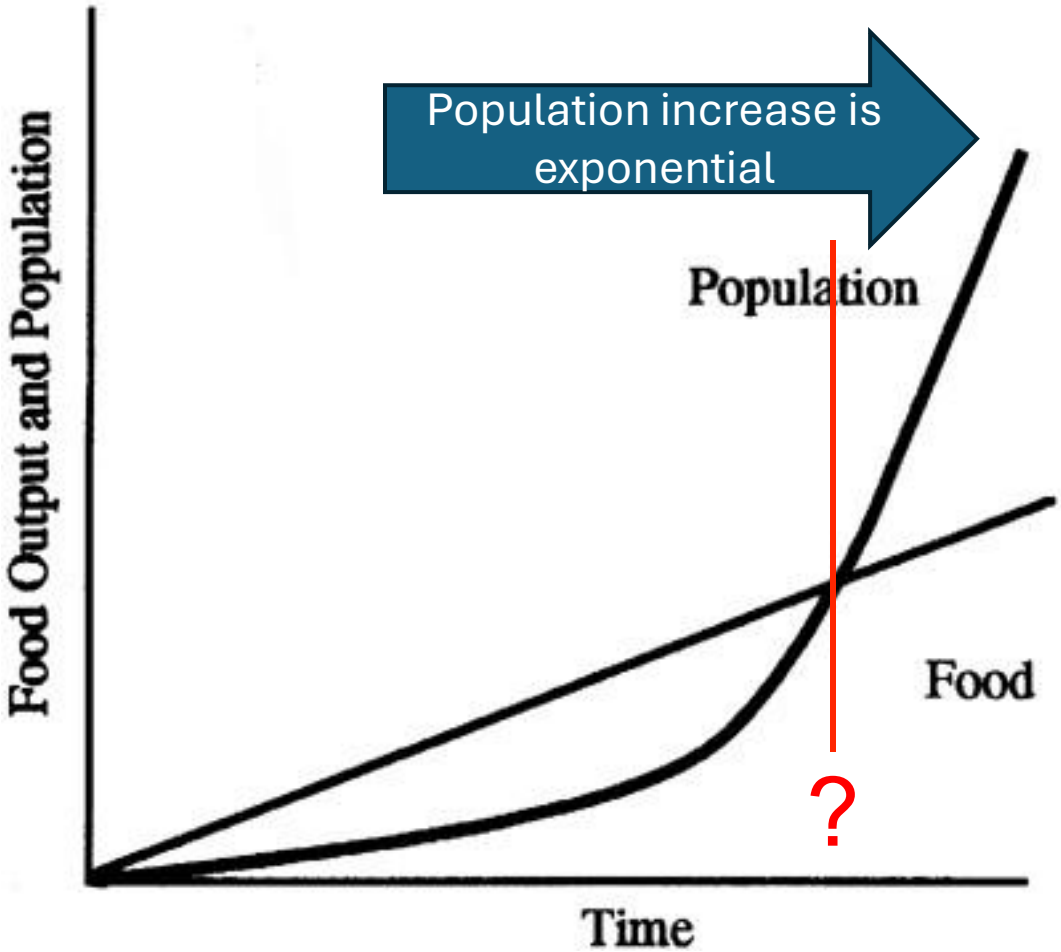
An Essay on the Principle of Population (1798)

Two fundamentals of population:

"First, That food is necessary to the existence of man."

"Secondly, That the passion between the sexes is necessary and will remain nearly in its present state."

“The power of population is indefinitely greater than the power in the earth to produce subsistence for man.” - Malthus



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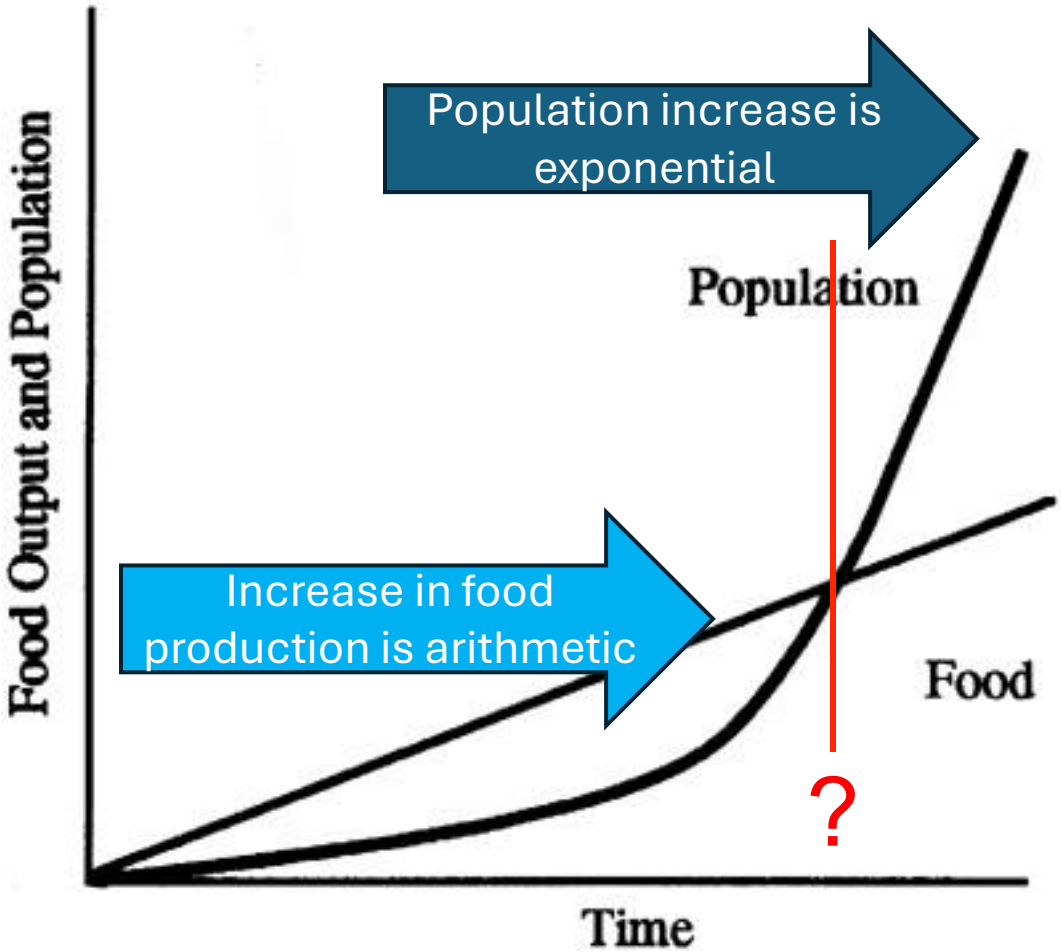

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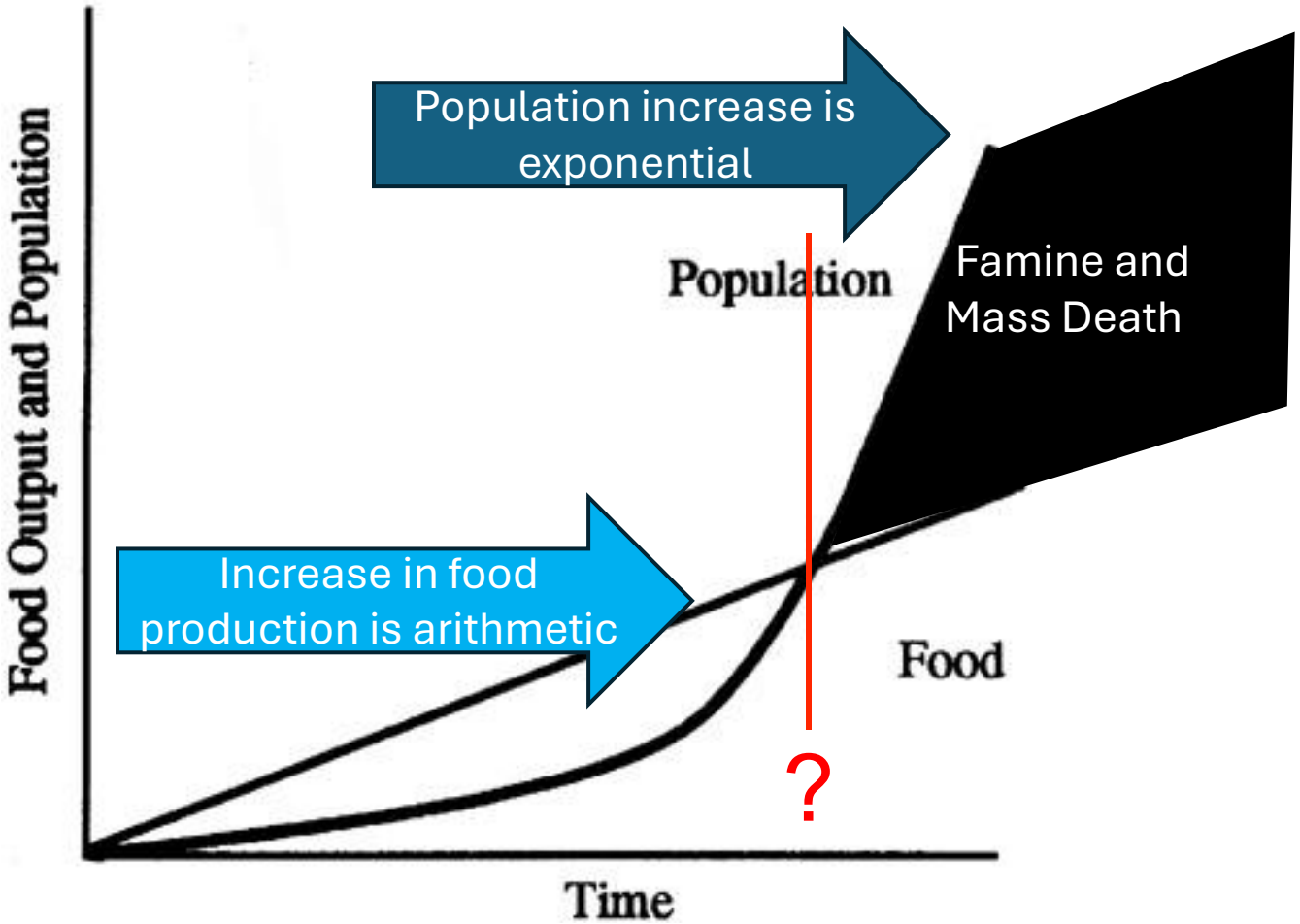


What did Malthus predict?

1. food production cannot keep up with population growth

2. population growth will inevitably produce poverty (cost of food would be too high for the poor)

3. the only preventative measure is ″moral restraint″

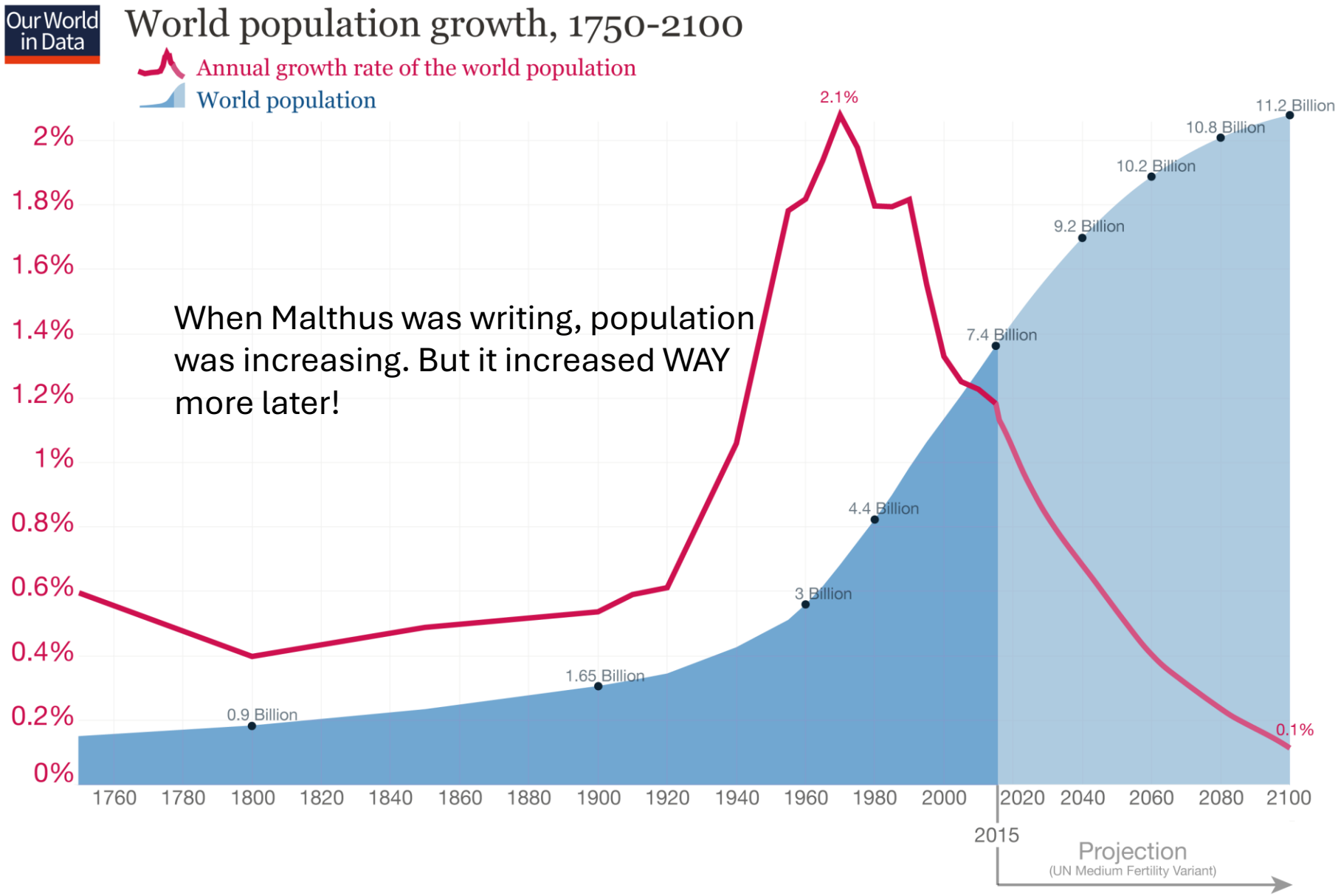
How did Malthus think this could be prevented?

- <u>Preventative checks (moral restraint)</u>
- Abstinence
- Delayed marriage
- Restrict marriage of the poor

- <u>Positive checks</u>
- Disease
- Starvation
- War

So, was he right?

- Lets look at some data.



When Malthus was writing, population was increasing. But it increased WAY more later!

Data sources: Up to 2015 OurWorldInData series based on UN and HYDE. Projections for 2015 to 2100: UN Population Division (2015) – Medium Variant.
The data visualization is taken from OurWorldinData.org. There you find the raw data and more visualizations on this topic.

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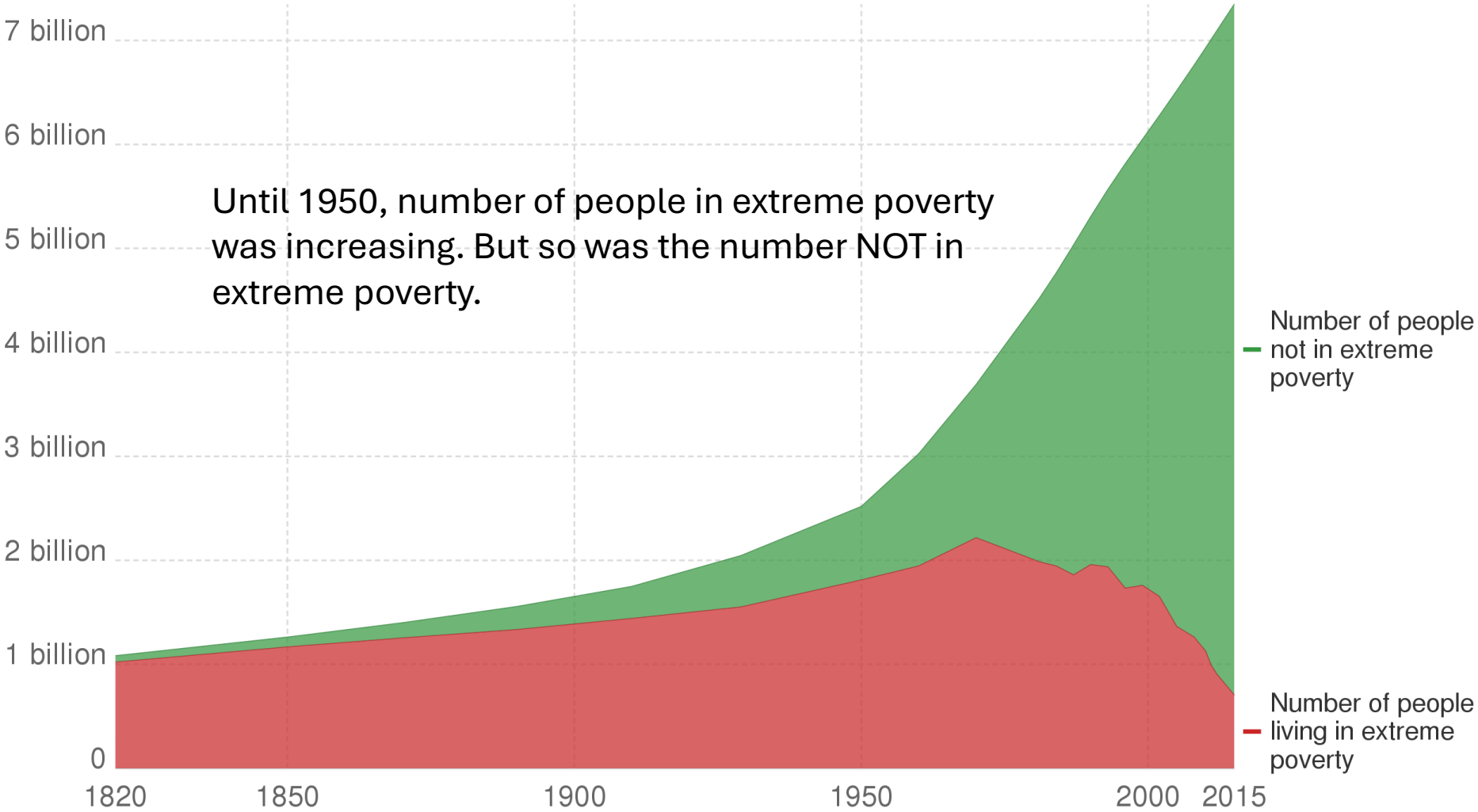
World population living in extreme poverty, 1820-2015

Extreme poverty is defined as living at a consumption (or income) level below 1.90 "international $" per day. International $ are adjusted for price differences between countries and for price changes over time (inflation).

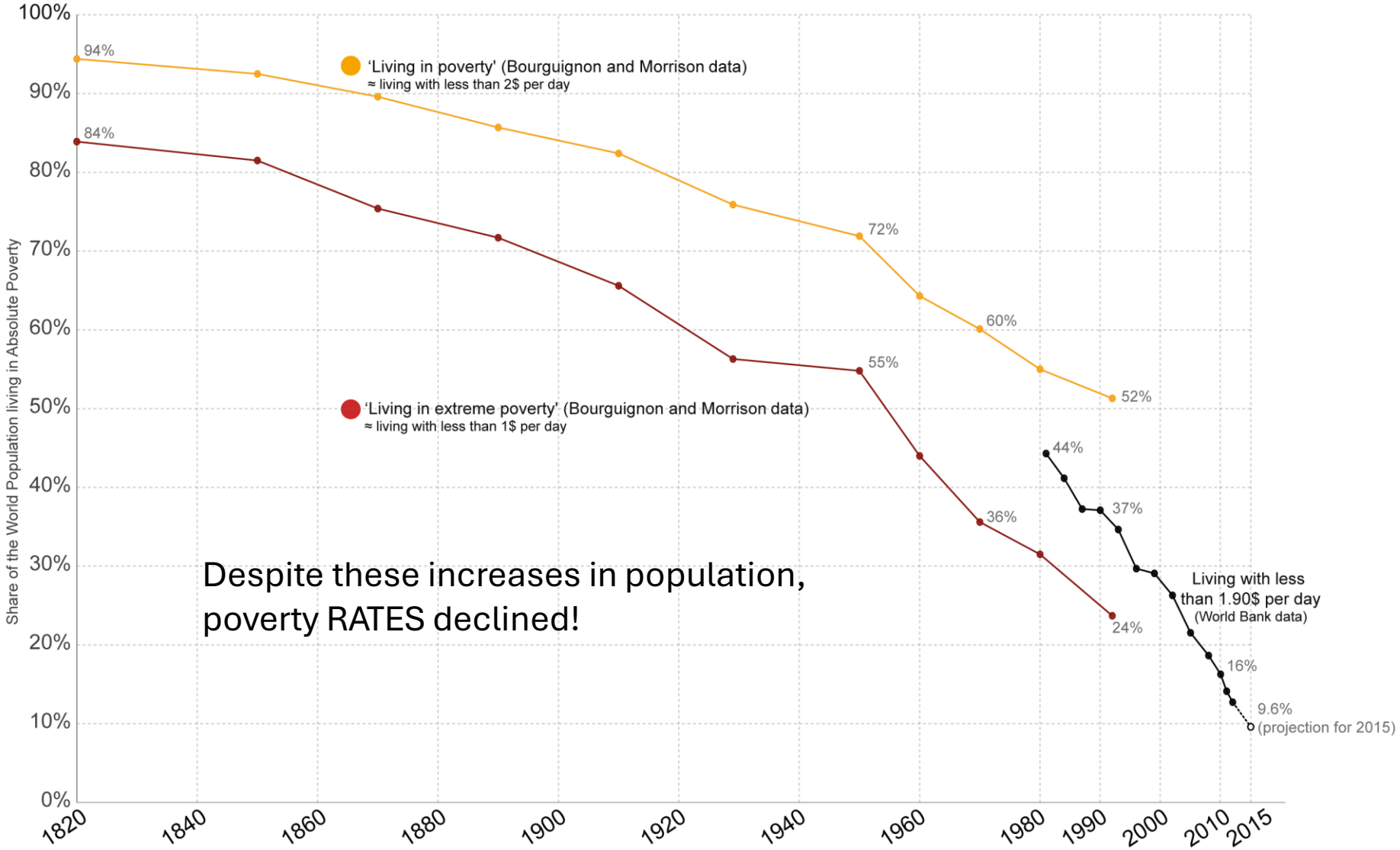


Source: World Poverty in absolute numbers - OWID based on World Bank (2016) and Bourguignon and Morrisson (2002)

Our World in Data

Share of the World Population living in Absolute Poverty, 1820-2015

All data are adjusted for inflation over time and for price differences between countries (PPP adjustment).



Data sources: 1820-1992 Bourguignon and Morrison (2002) - Inequality among World Citizens, In The American Economic Review; 1981-2015 World Bank (PovcalNet)

The interactive data visualisation is available at OurWorldinData.org. There you find the raw data and more visualisations on this topic.

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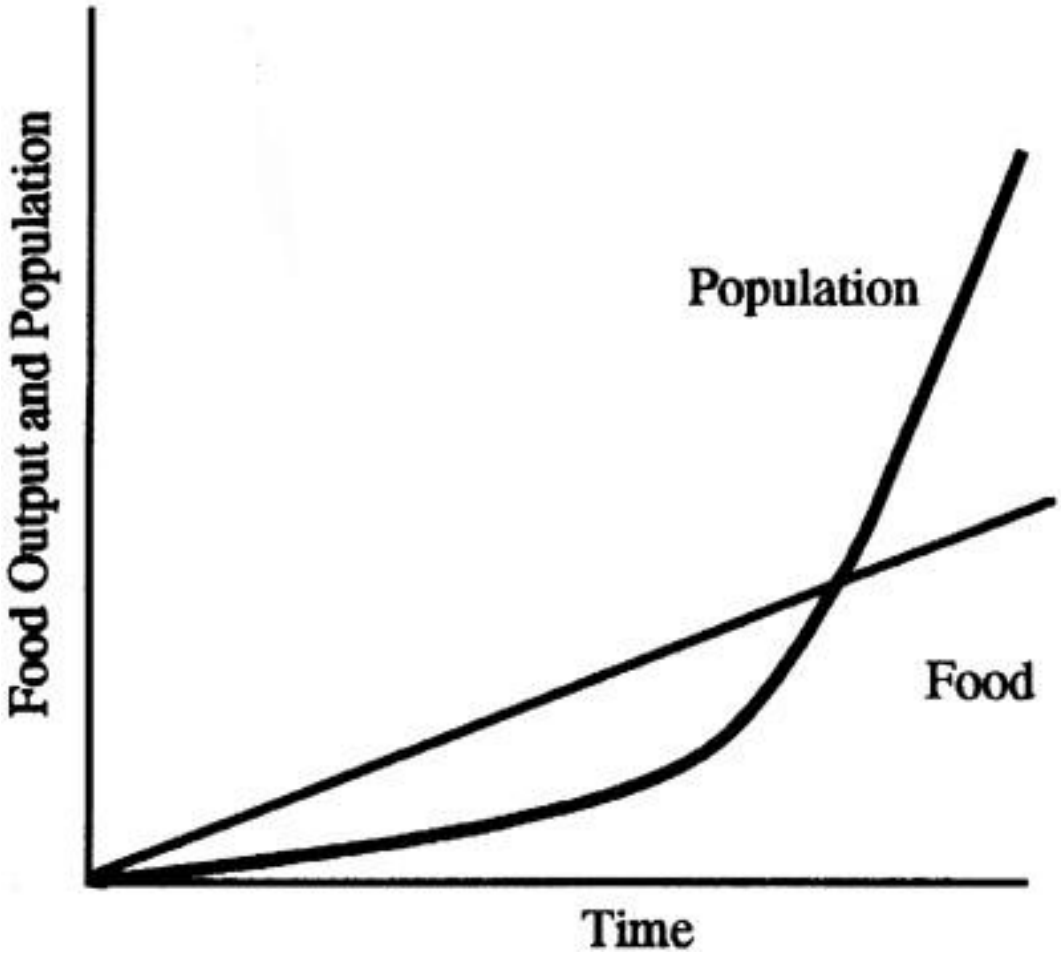
But, maybe we just haven't gotten there yet?



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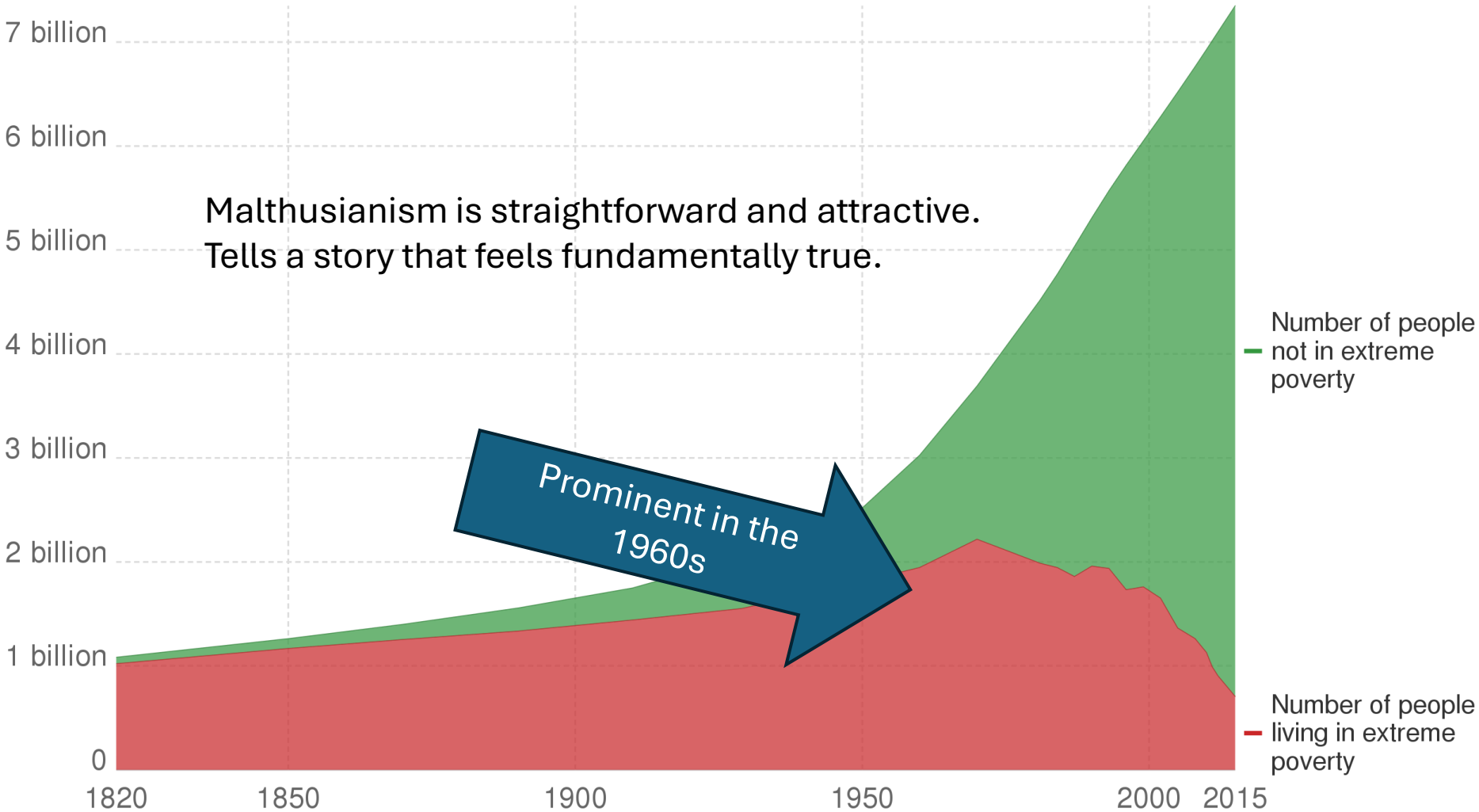

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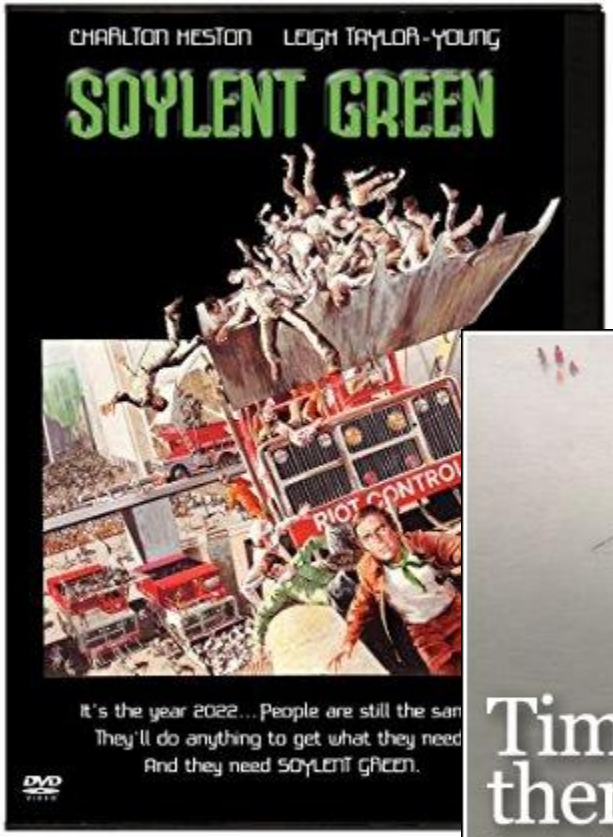
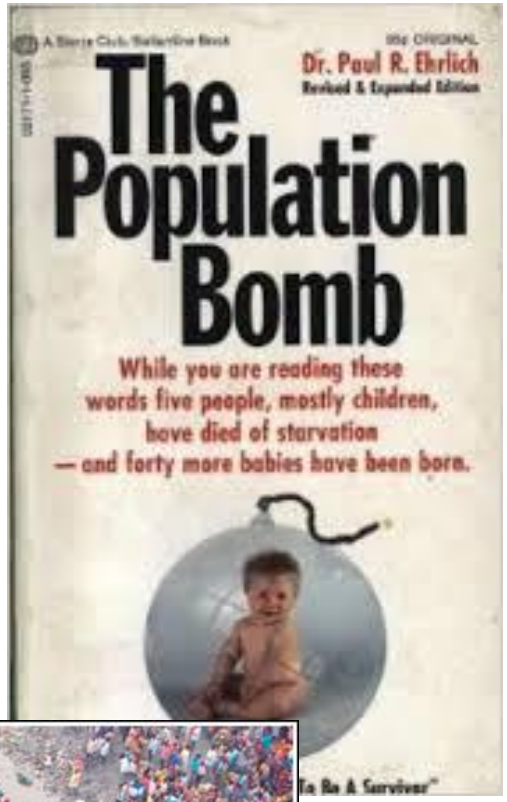
Neo-Malthusians





Critiques of Malthusianism

1. Moral
2. Technological
3. Demographic

Moral critiques of Malthusianism

- “Solving” overpopulation by coercion violates rights of least powerful

India 1970's

नसबन्दी सर्वोत्तम उपाय
STERILISATION IS BEST METHOD OF
FAMILY PLANNING
INCENTIVES FOR STERILISATION
2 CHILDREN Rs. 100/-, 3 CHILDREN Rs. 50/-
4 OR MORE CHILDREN Rs. 40/- (MALE) Rs. 20/- (FEMALE)

Table 1: Goals Achieved by Each Five-year Plan 1951–1979				
Five-Year Programs	**Goals in Terms of Family Planning**	**Allocation** (in millions of INR)*	**Expenditure** (in millions of INR)*	**Sterilizations****
First Five-Year Program (1951–1956)	Designed studies to determine reproductive patterns, provided information about birth control through existing hospitals, and constructed 147 new clinics	6.5	1.4	NA
Second Five-Year Program (1956–1961)	Constructed more than 4,000 clinics and propagated the "extension approach" i.e., benefits of having small families	49.7	21.5	152,677
Third Five-Year Program (1961–1966)	Failed to increase the national income, performed 1.5 million sterilizations, inserted 800,000 IUDs and increased family planning centers to 11,474	269.8	248.6	1,373,166
The Plan Holiday (1966–1969)	Incentivized sterilization, IUD insertions reached four million, distributed 300 million condoms, over 26,000 family planning clinics, 9,000 hospitals and health care centers providing information about birth control	829.3	704.5	4,391,996
Fourth Five-Year Program (1969–1974)	Clinics increased to 45,000, 862 mobile units, overall pace of family planning program reduced due to international oil crisis and Bangladesh War	2,857.6	2,844.4	9,003,626
Fifth Five-Year Program (1974–1979)	Voluntary sterilization changed to forced stelilization, more than 8 million sterilized.	NA	NA	18,500,000

*INR is India's rupee currency. For perspective, US $1 dollar equaled about 4 INR from 1950 to 1966, then the rupee began to devalue against the dollar over time (about 7.47 INR by 1970, 8.41 INR by 1975, and 10.18 in 1980 per US $1 dollar).

**Total number of male and female sterilizations.

Source: Kaval Gulati, "Compulsory Sterilization: The Change in India's Population Policy," *Science, New Series* 195, no. 4284 (1977): 1300-1305.

Source: Gupte 2017

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Voluntary sterilization changed to forced

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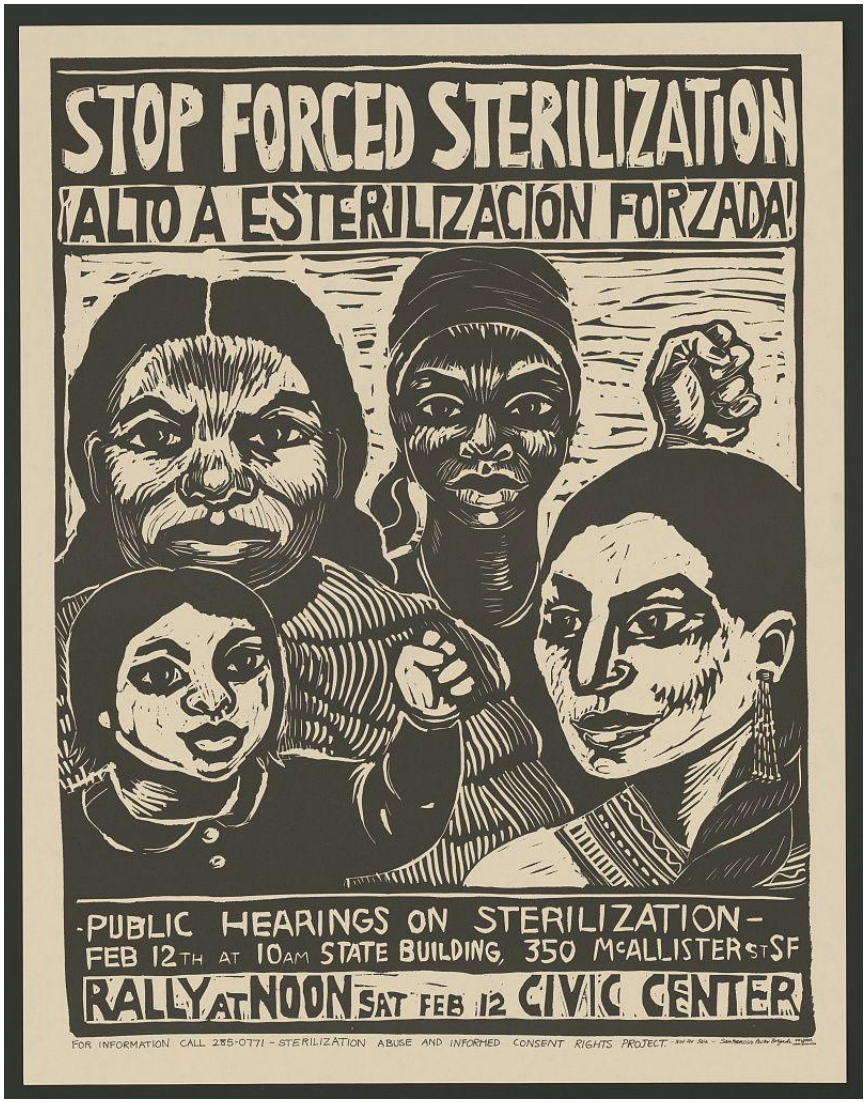


Puerto Rico (1960's)

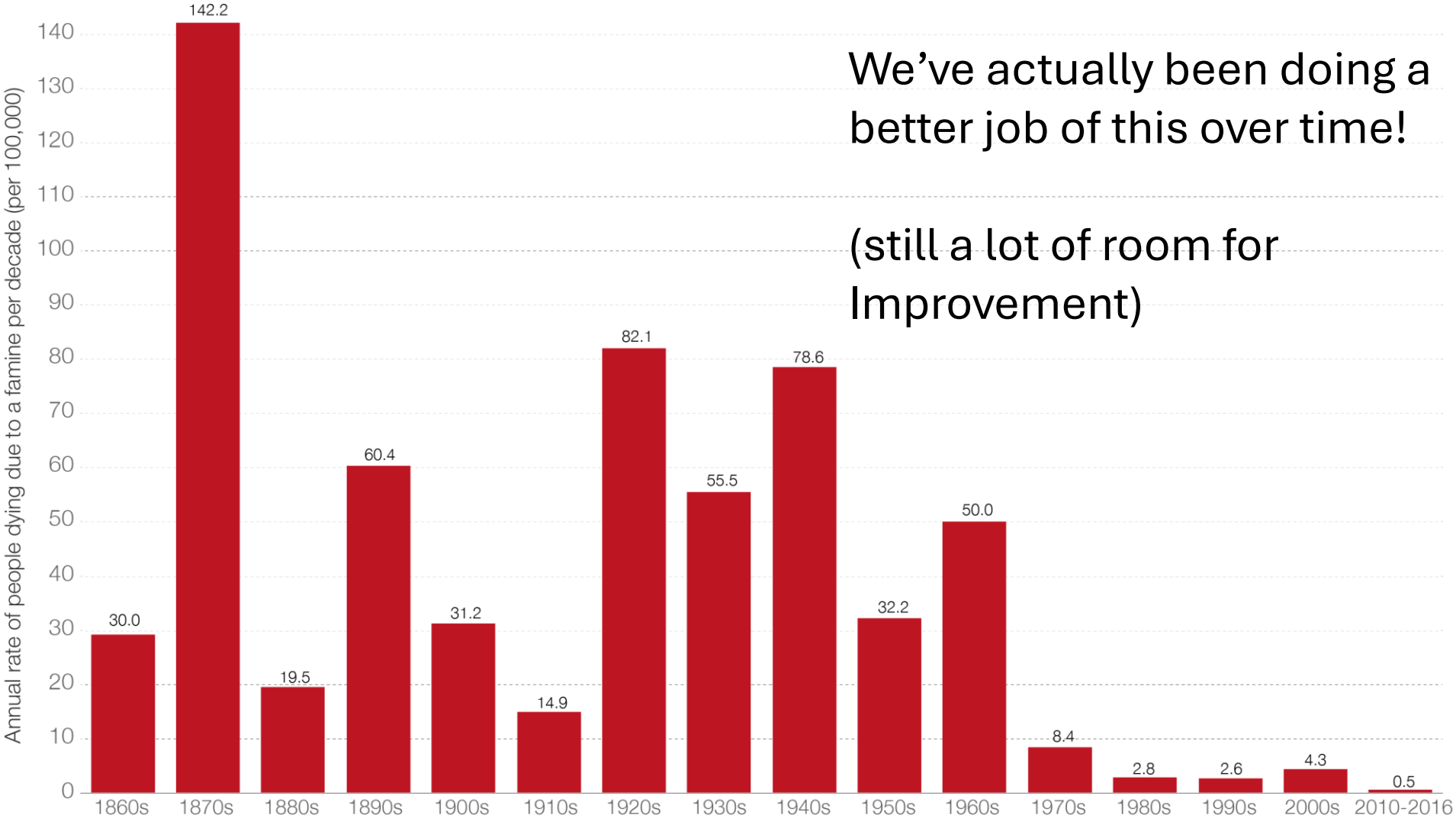
~1/3 of women sterilized without being told it was irreversible

Moral critiques of Malthusianism

- “Solving” overpopulation by coercion violates rights of least powerful
- Food shortages are problems of *access*, not population
  - There IS enough food to go around
  - Its just wasted or not in the right places

Annual rate of people dying due to a famine globally, per decade

Our World in Data



We've actually been doing a better job of this over time!

(still a lot of room for Improvement)

Data source: The rate of the excess mortality due to famines shown here is presented in detail on OurWorldinData.org/famines [The dataset was constructed by Joe Hasell and Max Roser].
How the rate is calculated: The annual rate is calculated for each decade by dividing the total number of famine victims in each decade by the average size of the world population in that decade and dividing the resulting ratio by 10 (and for the data from 2010 to 2016 it is divided by 7) to arrive at a yearly rate. To express it as the rate per 100,000 people, this annual rate is then multiplied by 100,000.
For famines that happened at the end of a decade and the beginning of the next decade the famine victims are split by decade on a year by year basis.
For famines for which different excess mortality estimates are published the midpoint between these estimates was chosen here.
This visualization is available at OurWorldinData.org. There you find the full dataset and more research and visualizations on famines and global development.

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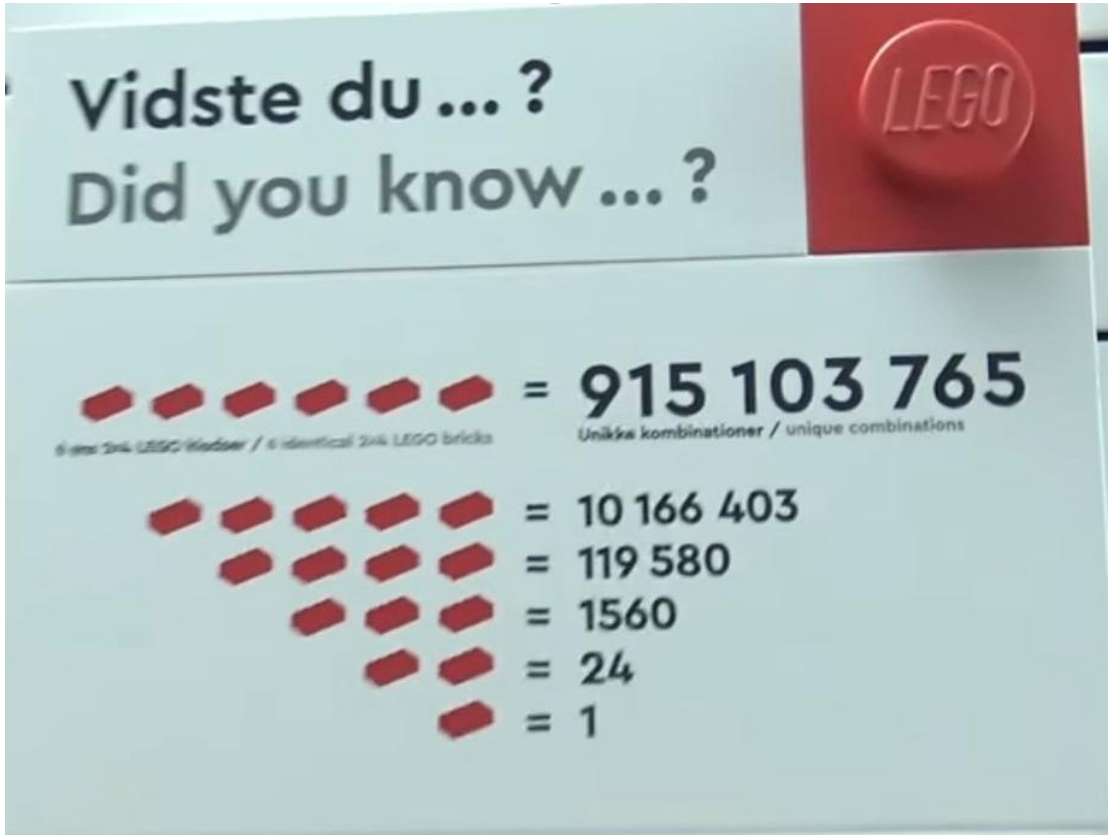
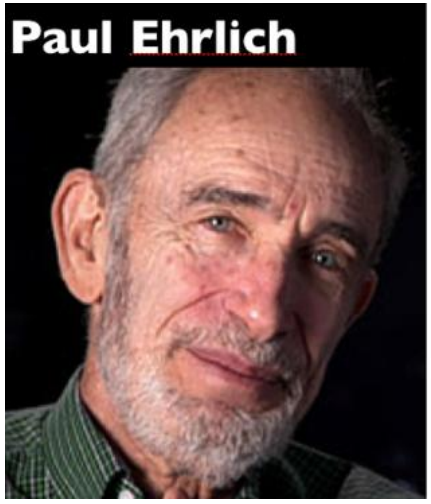
Technological critique of Malthusianism

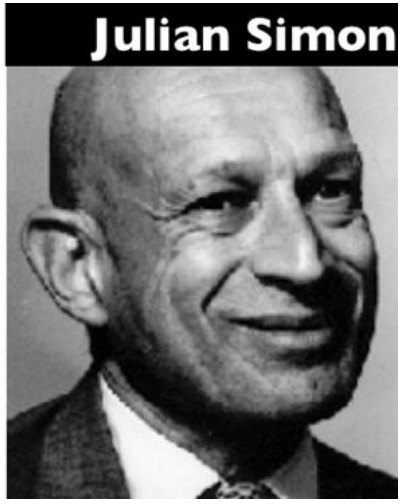
"Resource scarcity increases the cost of a product, thus creating incentive to find alternative materials."

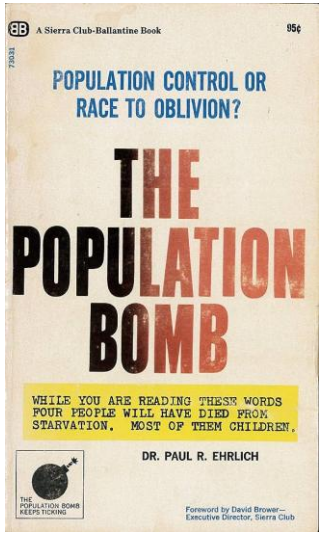
- Julian Simon (1981)

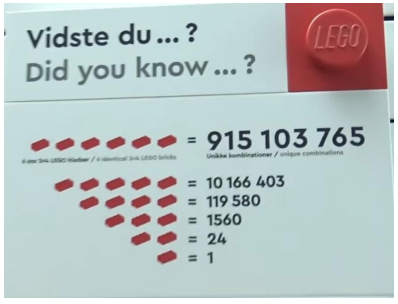

Vidste du ... ?
Did you know ... ?
LEGO
= 915 103 765
Unikke kombinationer / unique combinations
= 10 166 403
= 119 580
= 1560
= 24
= 1

Ehrlich v. Simon









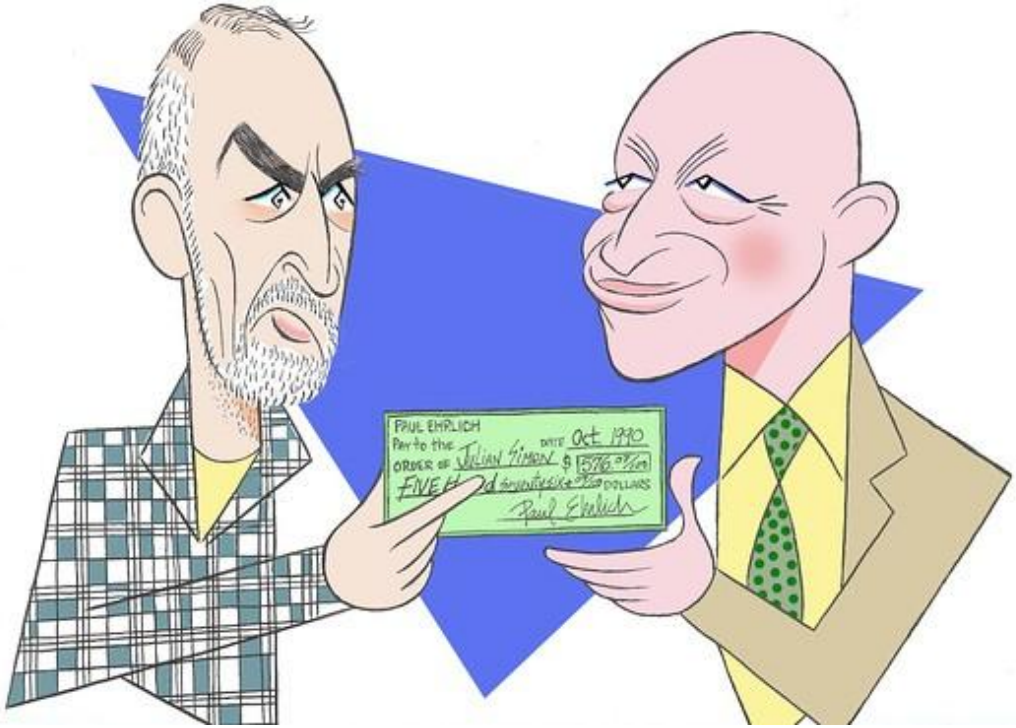
"World population has passed food supply. The famine has started."

- Paul Erlich (1965)

"Resource scarcity increases the cost of a product, thus creating incentive to find alternative materials."

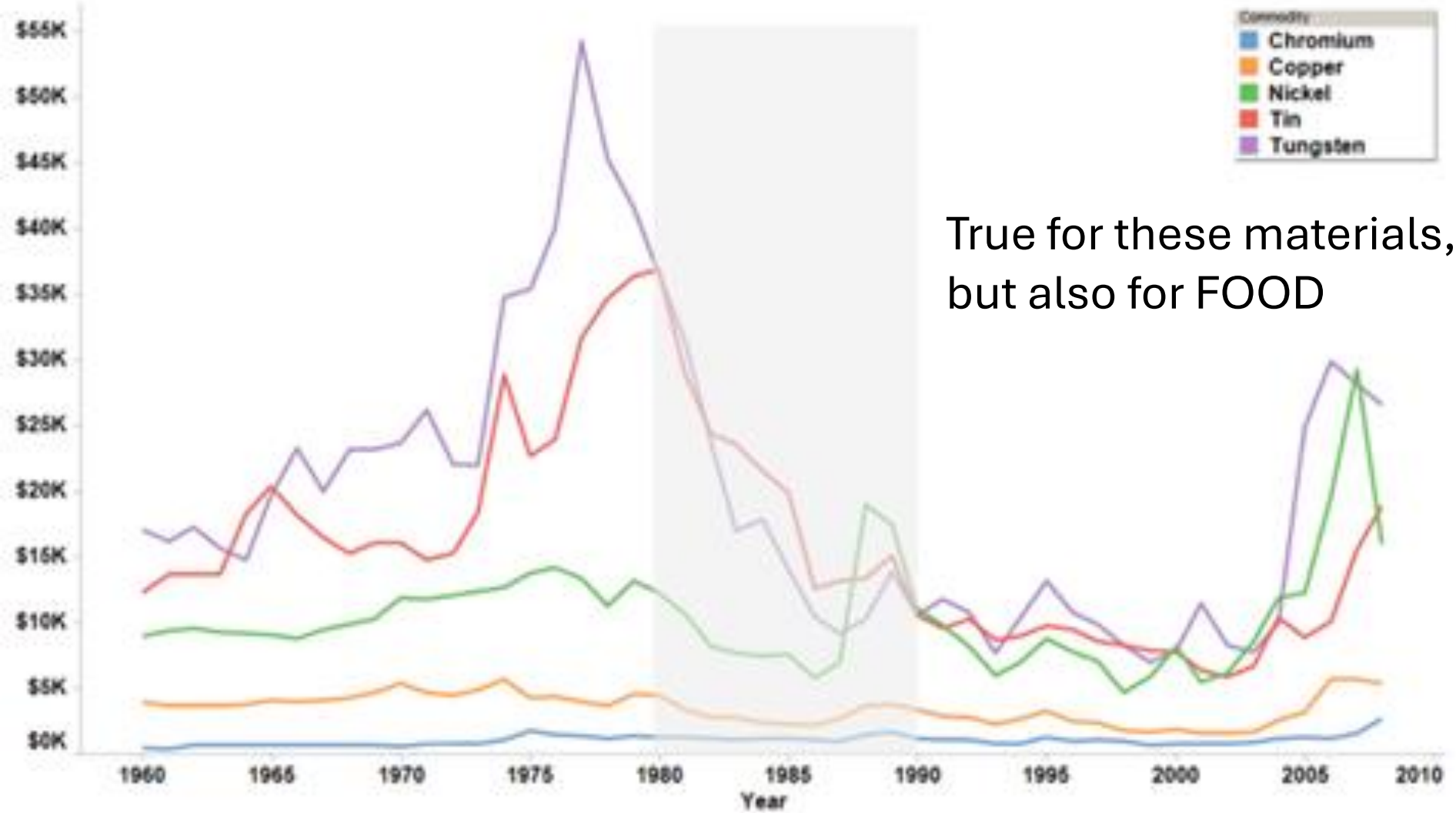
- Julian Simon (1981)

Ehrlich-Simon Bet (1980)



Simon offered to:

“Stake $10,000 ... on my belief that the cost of non-government-controlled raw materials will not rise in the long run."



True for these materials, but also for FOOD

“Green Revolution”

Norman Borlaug

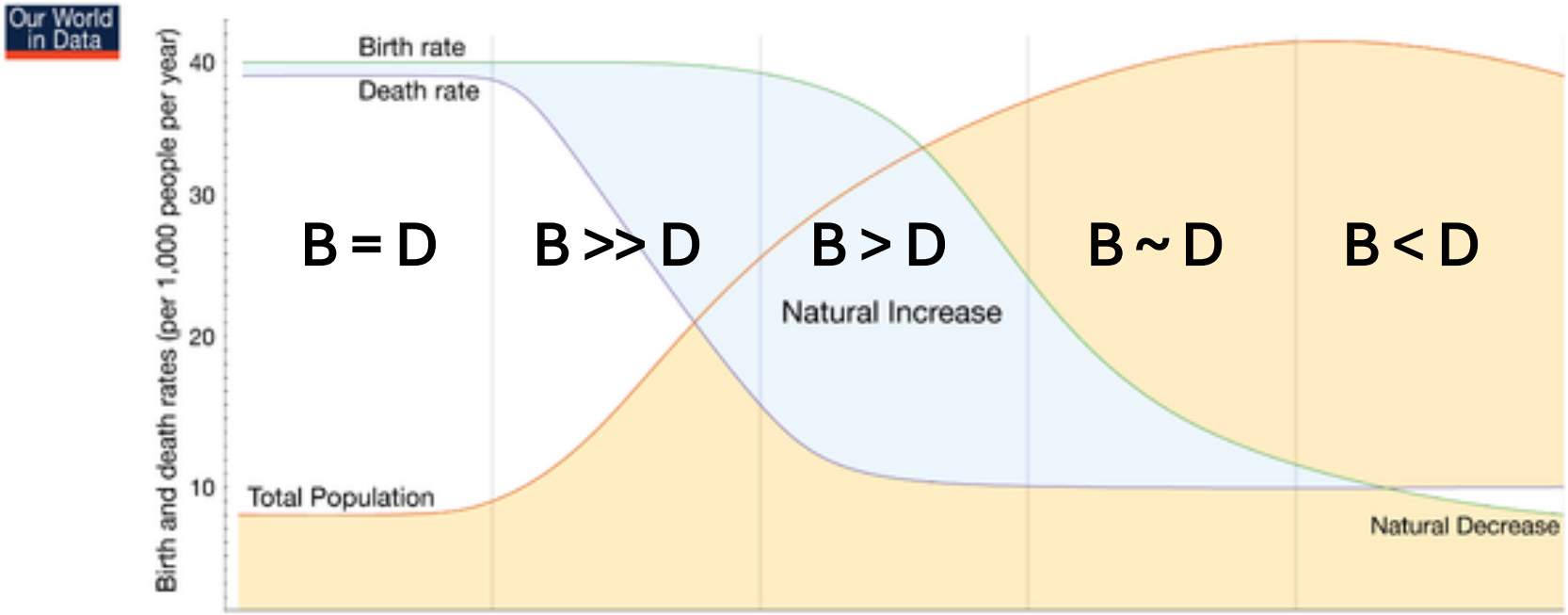
M.S. Swaminathan

Developed high-yield grain varieties that ~tripled food production

Demographic critique of Malthusianism

- Population growth rates vary over time as modernization progresses
- *Demographic transition* = generic model of sequential changes in mortality and fertility rates over time

The Demographic Transition



	Stage 1	Stage 2	Stage 3	Stage 4	Stage 5
Birth rate	*High*	*High*	*Falling*	*Low*	*Very low*
Death rate	*High*	*Falls rapidly*	*Falls more slowly*	*Low*	*Low*
Natural increase	*Stable or slow increase*	*Very rapid increase*	*Increase slows down*	*Stable or slow increase*	*Stable or slow decrease*

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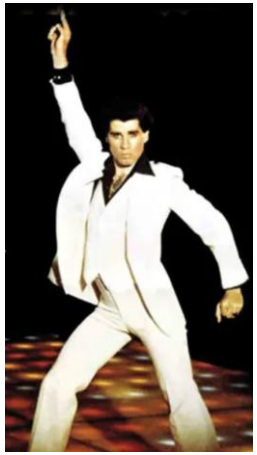
Malthusianism stays alive



- Simplistic story of Malthusianism is attractive.
- But not always right – often much more about consumption than sheer number of ppl
- Doesn't stop people from making the simplistic arguments!

SCIENTIFIC AMERICAN

ANTHROPOLOGY

Eight Billion People in the World Is a Crisis, Not an Achievement

More people will not solve the problem of too many people

By Naomi Oreskes on March 1, 2023

But…Elon Musk?

Critiques of Population Doomer-ism

1. Moral
2. Technological
3. Demographic

Critiques of Population Doomer-ism

1. Moral
   - Obvious: Forcing ppl to have babies is bad.
2. Technological
   - What would improved access to IVF, IUI and other assistive reproductive technologies mean?
   - Increased support for childcare?
3. Demographic
   - If you take any trend and extrapolate to infinity you get a doomsday scenario

Critiques of Population Doomer-ism

1. Moral
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2. Technological
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3. Demographic

Why do populations continue to grow when the fertility rate declines?

- **Population momentum**
  - Population growth continues even after the birth rates fall
  - When a large proportion of women are in the childbearing years, the total **number** of births can increase even when the **rate** of childbearing per woman falls
- **Time lag** – momentum keeps populations growing for many years after fertility declines
  - **Implication –** fertility changes now don't have big effects for a long time.

Population Momentum Example

Age	Population in Time 1	Population in Time 2
Babies	100	
Young People (can have babies)	2	
Old People (can't have babies)	2	

<u>TIME 1:</u>

Birth RATE? 100/2 = 50

Every young person in this population had 50 babies.

Population? 100+2+2 = 104

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Population? 100+2+2 = 104

TIME 2:

Birth RATE? 100/100 = 1

Every young person in this population had 1 baby.

Population?

Population Momentum Example

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Population? 100+2+2 = 104

HUGE decline in the birth RATE!

TIME 2:

E? 100/100 = 1

Every young person in this population had 1 baby.

Population?

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Population? 100+2+2 = 104

TIME 2:

Birth RATE? 100/100 = 1

Every young person in this population had 1 baby.

Population? 100+100+2 = 202

Population Momentum Example

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TIME 1:

Birth RATE? 100/2 = 50

Every young person in this population had 50 babies.

Population? 100+2+2 = 104

HUGE decline in the birth RATE!

HUGE increase in the population!

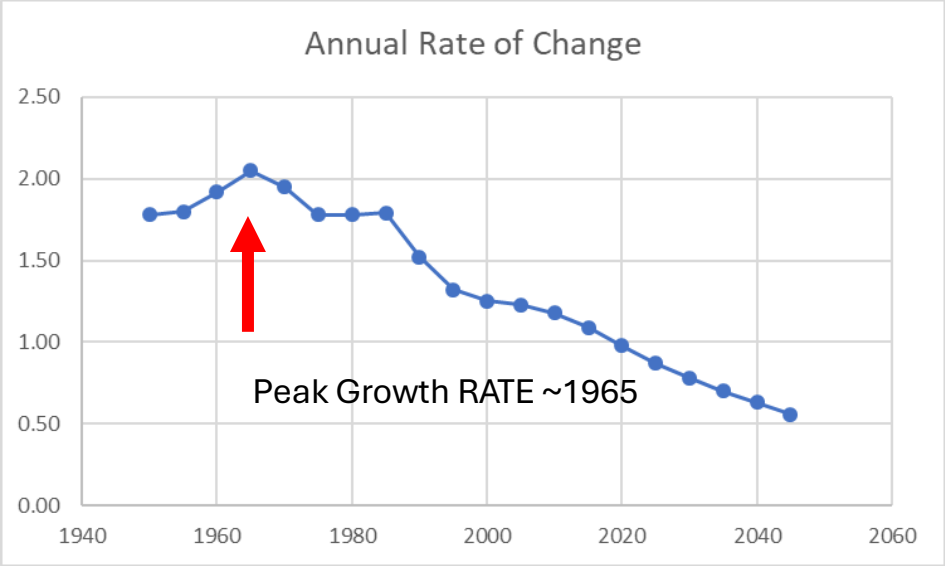
TIME 2:

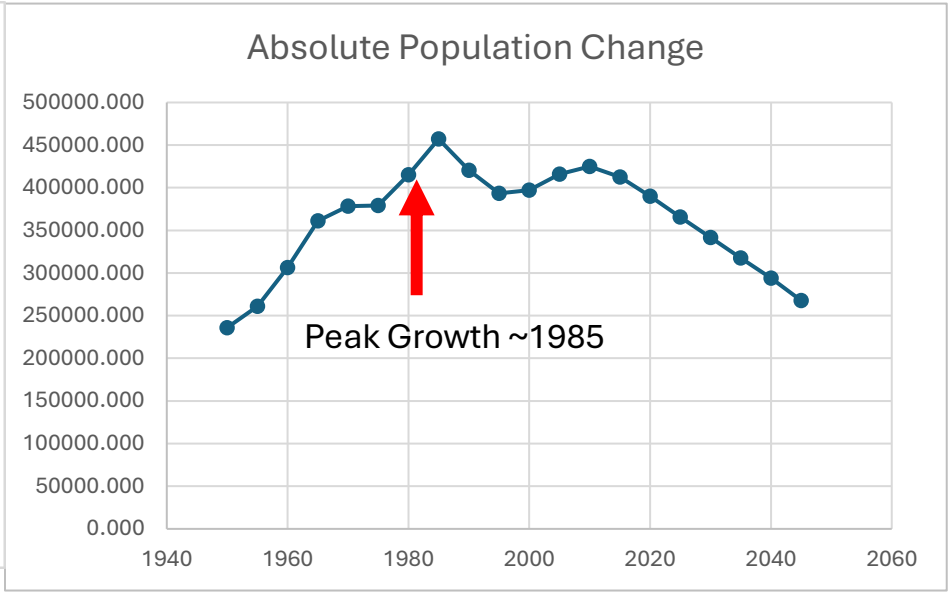
E? 100/100 = 1

Every young person in this population had 1 baby.

00+100+2 = 202

Global Population Momentum



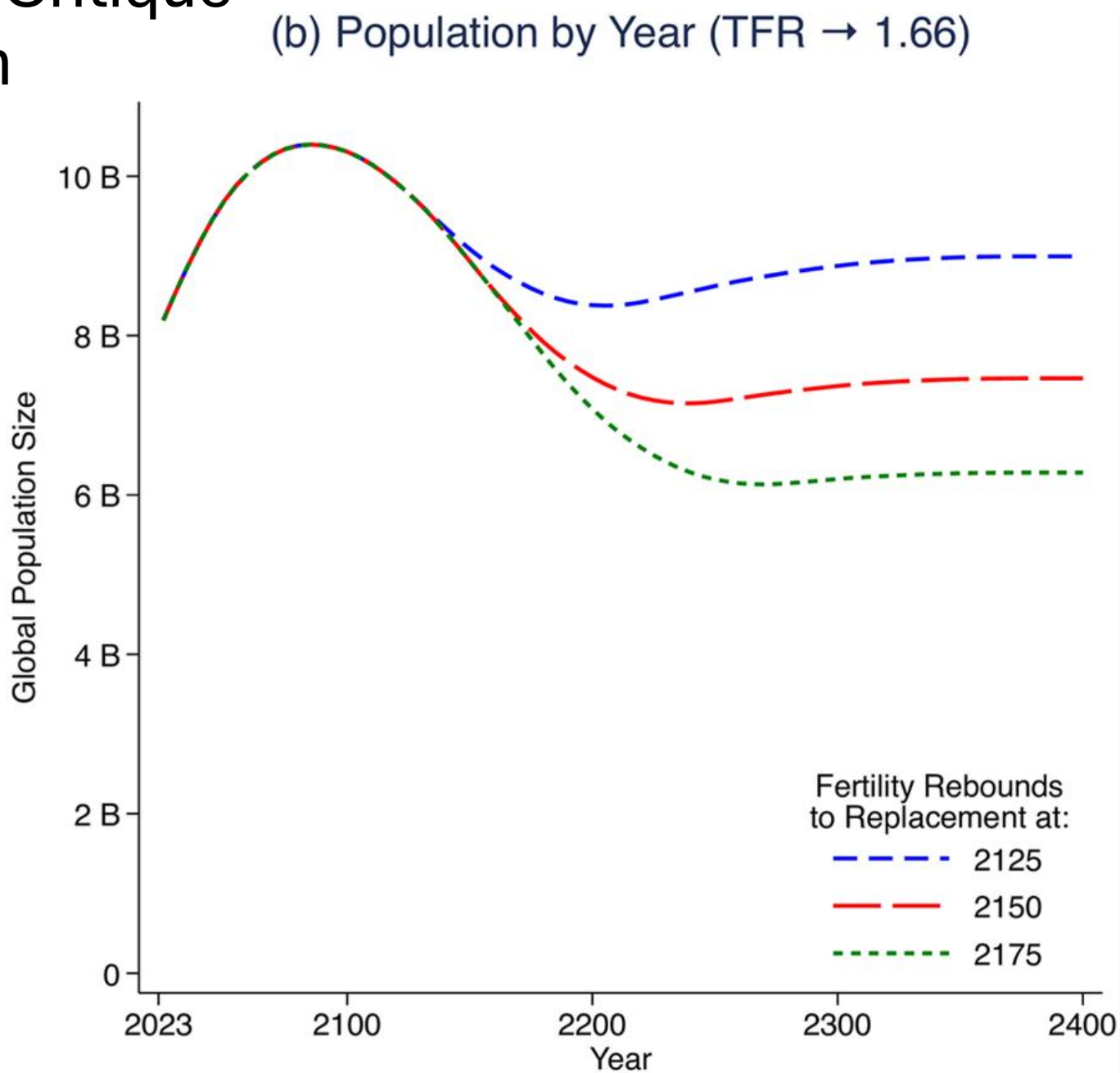


The rate of global population growth peaked in **early 1960s**, but the largest number of people was added to the world population in **late 1980s**

Demographic Critique of Doomerism

It will take a LONG time for global population to decline at current rates.

Why worry about what generations who wont be born for hundreds of years will do?



Conclusions

- Be wary of anyone making big claims about population, especially in the long term!
- **Advice** – don’t be fooled by extrapolations to infinity!

Question of the day

Was Malthus right? Explain in 1 sentence.

https://tinyurl.com/170soc0203